

Quiz 5 Problem Set

Quiz 5 will be based on these parts of the textbook:

- Chapter 13: 13.1 – 13.6
- Chapter 12: 12.1, 12.3

The questions on the quiz may refer to definitions, theorems, and examples from the textbook. The questions could be of any type, and will be variations of questions below, or questions that are similar.

Important: *Treat these problem-sets as non-AI activities!* Turn off all AI support and try to figure them out yourself **by hand**. Having AI or another student do this for you will not help you learn. You must do the learning yourself!

Question 1

Give the mathematical definitions of these terms: **graph**, **undirected graph**, **directed graph (digraph)**, **simple graph**, vertex **in-degree** and vertex **out-degree**, **path**, **cycle**, **subgraph**, **spanning subgraph**, **tree**, **spanning tree**.

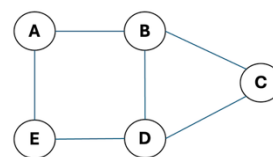
Question 2

Give mathematical justifications of these facts:

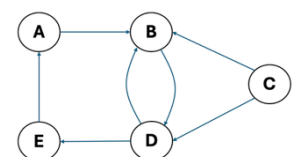
- The sum of the degrees of a (simple) graph is twice the number of edges. (Proposition 13.6)
- For a directed graph, the sum of all vertex in-degrees equals the sum of all vertex out-degrees. (Proposition 13.7)
- A tree with n vertices has exactly $n - 1$ edges. (Proposition 13.11)

Question 3

- Give an adjacency list for **graph 3.1**.
- Give an adjacency list for **graph 3.2**.
- Give an adjacency matrix for **graph 3.1**.
- Give an adjacency matrix for **graph 3.2**.



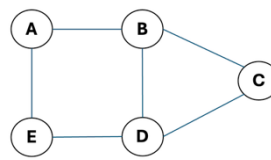
graph 3.1



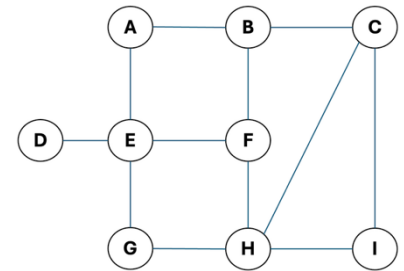
graph 3.2

Question 4

In the following questions, whenever there is a choice of what vertex to choose next, always choose the one that comes next alphabetically.



graph 4.1

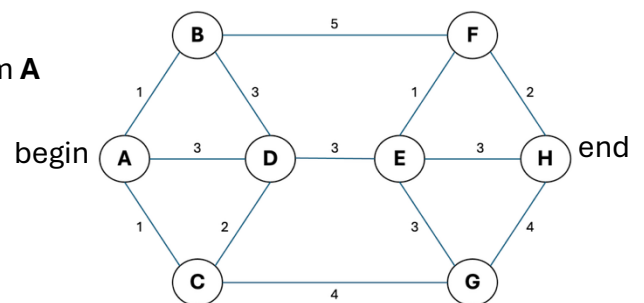


graph 4.2

- Draw the discovery edges and back edges that result from running **depth-first search** (DFS) on **graph 4.1**, starting at node **A**.
- Draw the discovery edges and back edges that result from running **depth-first search** (DFS) on **graph 4.2**, starting at node **A**.
- Draw the discovery edges and cross edges that result from running **breadth-first search** (BFS) on **graph 4.1**, start at node **A**.
- Draw the discovery edges and cross edges that result from running **breadth-first search** (BFS) on **graph 4.2**, start at node **A**.

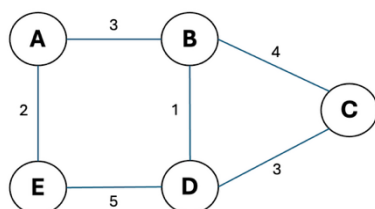
Question 5

Using Dijkstra's algorithm, give a shortest path from **A** to **H**, and also its weight.

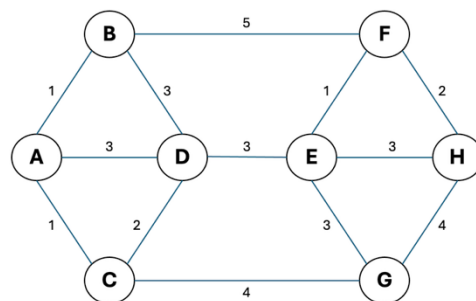


Question 6

Using **Kruskal's algorithm**, find a minimum spanning tree of graphs 6.1 and 6.2. If there is more than one minimal spanning tree for a graph, then give (at least!) 2 different trees.



graph 6.1



graph 4.2

Question 7

Explain the Boyer-Moore string matching algorithm, and the two main heuristics it uses to match strings.

Question 8

Show how to use **dynamic programming** to write functions that efficiently calculate the following:

- a) The Fibonacci numbers.
- b) Given a set of coin denominations as input, and a target number, finding that minimum number of coins that sum to the target.